👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 · STANDARD 6.NOS.7

Ordered Pairs in All Four Quadrants

Lesson 7-8 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can plot ordered pairs in all four quadrants of the coordinate plane.

Language Objective: I can explain my work using the words quadrant, negative coordinate, reflection, and axis.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Quadrant · **Negative coordinate** · **Reflection** · **Axis** · **Integer**

Level 3: try the blanks from memory first, then check the bank.

1 One of the four regions of the coordinate plane, numbered I to IV, is a _____.

2 A coordinate less than zero, which places a point left of or below the origin, is a _____.

3 A flip of a point over an axis is a _____ .

4 One of the two number lines, the x-axis or y-axis, that form the grid is an _____ .

5 A whole number or its opposite is an _____ .

Write About the Math

How do four-quadrant coordinates help a city map locations in every direction from the center?
¿Cómo ayudan las coordenadas de cuatro cuadrantes a una ciudad a ubicar lugares en todas las direcciones desde el centro?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Quadrant** *Cuadrante*
- ☐ **Negative coordinate** *Coordenada negativa*
- ☐ **Reflection** *Reflexión*
- ☐ **Axis** *Eje*
- ☐ **Integer** *Número entero*

Model: *The signs of the coordinates tell me the quadrant because ___.* — Student matches sign combinations to quadrants (e.g., (+,+) is I, (-,+) is II).

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The library is in Quadrant ___ because its x is ___ and its y is ___. The park is in Quadrant ___ because ___. Four-quadrant coordinates help because ___.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Quadrant
2. **Notes 2:** Negative coordinate
3. **Notes 3:** Reflection
4. **Notes 4:** Axis
5. **Notes 5:** Integer
6. **Watch for this mistake:** *A common mistake in Ordered Pairs in All Four Quadrants is matching each sign to the wrong axis — for example, seeing (-2, 5) and thinking "the x is negative so I move DOWN, and the y is positive so I move RIGHT," which lands the point in*

Quadrant IV. But the x-coordinate always controls left/right, never up/down: negative x means LEFT (not down), and positive y means UP (not right). So (-2, 5) is actually left 2, up 5 — Quadrant II. Before you submit, check that you used the x-coordinate for left/right and the y-coordinate for up/down, and never swapped them.

7. **Try It 1:** D. Quadrant III — *A negative x means left and a negative y means down, so (-5, -2) is in Quadrant III, where both coordinates are negative.*
8. **Try It 2:** A. (-, +) — *Quadrant II is up and to the left of the origin, so x is negative and y is positive: (-, +).*